René Doháň

Senior Software Engineer & Product Lead

Technology & Skills Profile

20+ years delivering software products, with senior expertise in full-stack engineering, native Android and iOS, C++ audio, and applied AI. Owns architecture, implementation, integration and production delivery across client products and an independent music-software platform.

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Proficiency guide

Self-assessed expertise and approximate years of development and ongoing maintenance, with overlapping projects counted once. Levels describe demonstrated engineering responsibility, from independent delivery through architecture and technical ownership.

Expert	Sustained architectural ownership and deep troubleshooting across substantial work.
Advanced	Independently designs, delivers and integrates complex solutions; resolves demanding technical problems.
Proficient	Independently delivers implementation and maintenance in established workflows.
Familiar	Limited practical exposure; uses reference material or guidance.

Programming languages

Technology	Level	Years (approx.)	Practical experience
Kotlin	Expert	10	Shared Android frameworks, music products, Motorkari, census applications and native SDK integration.
Java	Expert	20	Android applications and reusable libraries; earlier Swing, Eclipse RCP and enterprise Java products.
C++	Expert	6	Architecture and implementation of a shared real-time audio engine, transport, DSP and JNI integration.
Swift	Expert	11	Native iOS product ownership, reusable frameworks and SDK integration across Motorkari, the national census and Biolinq.
Objective-C	Expert	14	Architecture, delivery and long-term ownership of iPhone/iPad products and shared libraries, including Rcherz.
TypeScript	Advanced	7	Typed Vue/React products, Coinomatic services and infrastructure, localized content and build tooling.
JavaScript	Advanced	12	Node.js services, Firebase generation workers and web applications; earlier GWT/PHP product integrations.
PHP	Advanced	5	Historical SaaS and PDF-processing work at QualityUnit, plus focused PHP/Yii product changes for Plaut.
C# / .NET	Proficient	4	Earlier desktop/OCR applications and coordinated .NET GraphQL API changes for Sensoneo.

Programming languages

Technology	Level	Years (approx.)	Practical experience
Python	Proficient	2	Android capture/reporting tools, asset workflows and music-sheet utilities.
Shell / Bash	Advanced	7	Repeatable Android builds, release automation, native dependency builds and cloud operations scripts.
SQL	Advanced	10	Relational data changes and queries across PostgreSQL, MySQL and mobile SQLite workflows.

Mobile & native platforms

Technology	Level	Years (approx.)	Practical experience
Android SDK / AndroidX	Expert	15	Production Android products, lifecycle/state architecture, reusable UI libraries and platform modernization.
iOS SDK / UIKit	Expert	14	End-to-end native product engineering, architecture, device integration and reusable foundations across multiple released applications.
Android NDK / JNI	Advanced	6	Kotlin-to-C++ audio interfaces, native state and real-time callback boundaries.
Kotlin Coroutines	Advanced	5	Lifecycle-aware requests, asynchronous state, cancellation and shared Android permission handling.
Room / SQLite	Advanced	6	Offline data and structured local persistence, including national-census field workflows.
Gradle / Android Gradle Plugin	Advanced	12	Multi-module applications, product variants, native builds, dependency upgrades and release configuration.
Swift Package Manager	Advanced	11	Reusable Swift library packaging and dependency management across actively maintained codebases.
CocoaPods	Advanced	10	Dependency and build maintenance across established native iOS applications.
Mobile REST integration	Expert	15	Authenticated service integration, synchronization, pagination, failure recovery and offline behavior.
FCM / Amazon Device Messaging	Advanced	8	Push messaging, notification behavior and device/platform compatibility across client products.
Google Maps / MapKit	Advanced	8	Location permissions, map lifecycle, markers, clustering, routes and nearby-search workflows.
Bluetooth LE / BLE MIDI	Advanced	5	MIDI device discovery, connection lifecycle and input/output integration in music applications.
Media3 / ExoPlayer / Google Cast	Advanced	3	Media playback and video presentation, including playlists and full-screen behavior in Motorkari.

Mobile & native platforms

Technology	Level	Years (approx.)	Practical experience
Native SDK / API design	Expert	15	Reusable Android/iOS frameworks, public library interfaces, device SDK integration and consistent cross-platform behavior.
Google Play Billing	Advanced	5	Paid application variants, consumable credits, purchase verification, reconciliation and account adoption.

Audio, DSP & MIDI

Technology	Level	Years (approx.)	Practical experience
Real-time audio / DSP	Expert	6	Shared engine for effects, multitrack playback, recording, instruments, routing and low-latency performance.
MIDI / sequencing / synchronization	Expert	6	MIDI I/O, clock handling, event queues, controllers, musical sequences and transport synchronization.
Audio transport / recording / looping	Expert	6	Playback timing, seek/wrap recovery, overdubbing, crossfades and coordinated multitrack processing.
Oboe / AAudio	Advanced	4	Android low-latency stream integration, output reconnection, routing recovery and performance configuration.
FFmpeg / FFmpegKit	Proficient	3	Native dependency builds, audio conversion/decoding and validation of generated audio results.
Rubber Band	Proficient	1	Integration and build work for sample time-stretching and pitch-processing workflows.
libsndfile	Proficient	4	Audio-file I/O integration within the native music-software toolchain.
CMake	Advanced	5	Native audio-engine builds, Android ABI configuration and C++ optimization/build variants.

Web & backend

Technology	Level	Years (approx.)	Practical experience
React	Advanced	3	Full-stack React product delivery: application architecture, responsive workflows, service integrations, automated tests and production releases.
Next.js	Proficient	1	Marketing-to-application routing, release integration and Payload CMS work in existing Next.js products.
Vue.js / Vue 2 / Vue 3	Advanced	7	Client interfaces, the Renetik services portal and the music-product documentation website.

Web & backend

Technology	Level	Years (approx.)	Practical experience
VuePress 2 / Vite	Advanced	1	Product guides, custom components, structured Markdown, optimized assets and combined-site builds.
HTML / CSS / SCSS / Stylus	Advanced	12	Responsive product interfaces, themes, accessible content presentation and printable document layouts.
Vuetify / Vuex	Advanced	5	Component-based Vue products, shared presentation and application-state workflows.
Node.js	Advanced	5	Backend services, Firebase workers, agent runners, translation tools and build/release orchestration.
REST API design	Advanced	10	API architecture across web, backend and mobile products: stable contracts, authentication, validation, idempotency and integration clients.
Payload CMS / Lexical	Advanced	1	Restricted publishing identity, REST contracts, draft/publish safeguards and rich-text article integration.
Spring / Spring Boot	Advanced	7	Backend services and continued maintenance supporting Android field applications, data workflows and image exports.
GraphQL	Proficient	4	Client/API integration and ongoing use, including coordinated Kotlin and .NET product changes for Sensoneo.
Yii	Proficient	2	PHP product development and maintenance, including PostgreSQL-backed application work for Plaut.
Technical SEO / structured data	Advanced	1	Prerendering, canonical routes, localized metadata, sitemaps, redirects and release validation.

Cloud, data & operations

Technology	Level	Years (approx.)	Practical experience
AWS Lambda / API Gateway / Cognito	Advanced	1	Coinomatic APIs, identity, asynchronous runtime boundaries and authenticated product workflows.
SST / AWS CDK	Advanced	1	Infrastructure as code, stage configuration, runtime deployment and operational smoke checks.
EC2 / VPC / IAM / systemd	Advanced	1	Fixed-egress runtimes, host isolation, restricted permissions, agent scheduling and release operations.
Amazon SQS / asynchronous workers	Advanced	1	Durable dispatch, retry/dead-letter handling, idempotency and recoverable automation execution.
Amazon S3 / CloudFront	Advanced	3	Application delivery, exports and stored media; earlier field-app image export integration.

Cloud, data & operations

Technology	Level	Years (approx.)	Practical experience
CloudWatch / SNS / SES	Advanced	1	Operational logs, metrics, alarms, health checks, customer notifications and publishing reports.
DynamoDB	Advanced	1	Workspace-scoped application data, durable state, conditional updates and auditable automation records.
Firebase Auth / Firestore / Storage	Advanced	7	Music-product services, content delivery and authenticated data; recent generation/account isolation.
Firebase Crashlytics / Analytics	Advanced	7	Production diagnostics, product feedback and usage instrumentation across Android applications.
Cloud Functions for Firebase	Advanced	3	HTTPS APIs, account/credit services, generation dispatch, reconciliation and cleanup.
Google Cloud Tasks / Cloud Run	Advanced	1	Bounded generation queues, concurrency, retry controls and backend execution monitoring.
PostgreSQL / Supabase	Proficient	2	Relational application data, ongoing backend maintenance and Payload CMS database/migration integration.
MySQL	Proficient	3	Relational product data and coordinated backend changes, including Sensoneo.
Microsoft Azure	Proficient	1	Changes and integration within the Azure-hosted Sensoneo product.
Docker	Proficient	2	Container-based development and deployment workflows supporting maintained PHP/backend products.
Stripe	Advanced	1	Subscription billing, pricing, checkout, webhooks, refunds and event reconciliation for Coinomatic.
Vercel	Advanced	1	Production application/marketing integration and isolated CMS/website Preview releases.
GitHub Pages	Advanced	5	Static portal and product-site delivery, combined artifacts, compatibility redirects and production checks.

Applied AI & agentic systems

Technology	Level	Years (approx.)	Practical experience
ElevenLabs audio APIs	Advanced	1	Prompt-based music/audio generation, provider request mapping, queued execution and result validation.
DeepSeek / OpenRouter	Advanced	1	Musical-preset generation and provider-backed planner, writer and reviewer sessions.
LLM tool calling / JSON Schema	Advanced	1	Bounded model tools, structured responses, validation, corrective retries and token/request limits.

Applied AI & agentic systems

Technology	Level	Years (approx.)	Practical experience
Knowledge retrieval / BM25	Advanced	1	Versioned Markdown knowledge bundles, ranked section search and bounded read tools for grounded content.
Agentic workflow orchestration	Advanced	1	Planner/writer/factual-review/editorial-review pipeline, deterministic publication and verified replay in development.
AI job lifecycle / credit accounting	Advanced	1	Product-scoped accounts, transactional reservations/settlement, purchase verification, reconciliation and spend caps.
Codex / Claude Code / ChatGPT	Advanced	1	Specification-led research, implementation, review, testing and debugging with direct engineering accountability.
AI-assisted localization	Advanced	1	Multi-provider translation catalogs with source hashes, incremental updates, validation and generated locale assets.

Testing & delivery

Technology	Level	Years (approx.)	Practical experience
JUnit / Espresso / UI Automator	Advanced	8	Android regression checks, native integration scenarios, UI automation and repeatable screenshot capture.
Vitest / Playwright	Advanced	1	Coinomatic unit/integration, end-to-end and UI-regression testing with deployed smoke validation.
Jest / Puppeteer	Advanced	1	Portal model/component tests, prerendering, PDF generation and browser-based release checks.
Node.js test runner / Firebase emulators	Advanced	1	Generation lifecycle, account boundaries, provider behavior and storage-rule tests.
Gherkin / behavior-driven scenarios	Proficient	1	Native SDK behavior and early test scenarios in an API-first medical-device team.
Git / GitHub	Advanced	15	Independent repositories, shared libraries, branches, code reviews and release coordination.
CI/CD / Fastlane / R8	Advanced	6	Multi-variant Android releases, store metadata, build artifacts, code shrinking and automated deployment checks.
Scrum / TDD / product specification	Advanced	8	Team delivery, pair programming, requirements refinement, acceptance criteria and technical product ownership.

Earlier software & systems experience

Technology	Level	Years (approx.)	Practical experience
Java Swing	Advanced	4	Historical desktop applications and a magazine PDF viewer, including PHP integration.
Eclipse RCP	Advanced	3	Historical scripting IDE/editor with highlighting, formatting and script execution at QualityUnit.
J2EE / JSF	Proficient	3	Historical enterprise Java web-application development at QualityUnit.
GWT / Comet / MVP	Proficient	3	Historical LiveAgent SaaS interface, event-driven behavior and PHP/JavaScript integration.
PDF parsing/generation / OCR	Advanced	3	Historical PHP PDF parser/rebrander and C# batch document-recognition software.
Windows / PC hardware support	Proficient	5	Historical PC assembly, installation, servicing and technical customer support.
Linux / TCP/IP / wireless networks	Proficient	4	Earlier systems/network support and telecommunications installations; recent Linux cloud-host operations.
AutoCAD support	Familiar	1	Historical installation and user support for architectural computing; not architectural CAD design.